



Figure 3.5 Jenn Panko RAHT, CCRP (COCM) performing underwater treadmill therapy on a cat.

College of Veterinary Sports Medicine and Rehabilitation must evaluate the horse, the farrier, the saddle fit, and the role of the rider in causing or magnifying the lameness (Davies, 2014b). The rider may need to undergo their own rehabilitation program if problems are caused by the rider's posture and balance, and collaboration with a physical therapist or chiropractor treating the rider is beneficial. An entire chapter is devoted to equine rehabilitation (see Chapter 22).

Pain Assessment for Horses

Behavioral changes associated with pain in the horse may include kicking or biting when the tender area is touched, generalized restlessness, sweating, frequent movement of the painful limb, or continuous shifting of weight from one limb to another (Davies, 2014b). Changes associated with back pain, such as poor performance, poor appetite, and slight changes in demeanor may be very subtle initially. These may progress to tail swishing



Figure 3.6 Horse underwater treadmill at the University of Tennessee. *Source:* Courtesy of Dr. Steve Adair.

or holding the tail to one side, bruxism, head shaking, resistance to saddling and grooming, loss of flexibility, stumbling, bucking, and rearing (Marks, 2000; Paulekas and Haussler, 2009). The horse may be withdrawn and the abdomen may have tight abdominal muscles or appear to be “tucked-up.” The horse's breed and temperament must also be assessed.

Objective measurements of pain in the horse that may be of assistance in diagnosing and treating pain include heart rate (not consistently increased in pain), respiratory rate, pressure algometry, thermography, kinematic gait analysis, and response to analgesia (e.g., nerve blocks). Many times the horse is placed on analgesics (usually non-steroidal anti-inflammatory drugs (NSAIDs)) as a trial

to see if pain is present. If the problem does not improve with this single medication, then the horse is often incorrectly declared to not be in pain. This can lead to much more severe riding or controlling techniques (e.g., double twisted wire bit) being used on a horse in pain in order to “control” perceived bad behavior, when the problem is actually a normal response to pain.

The credentialed rehabilitation veterinary technician will be assisting the CERP veterinarian in carrying out the treatment plan for alleviation of this pain. Initially, the work is all on the ground to strengthen the abdominals and core musculature, followed by introduction of a rider (Davies, 2014b).

Ways to Treat Equine Pain

These include:

- analgesics,
- acupuncture,

- physical rehabilitation therapy (Figure 3.6),
- local heat,
- osteopathy,
- veterinary spinal manipulation or “veterinary chiropractic,” and
- massage.

Conclusion

The certified rehabilitation veterinarian, certified rehabilitation veterinary technician, physical therapist, and owner must work as a team to have successful management of pain and restore the patient (no matter what species) to a functional life. Each animal should have its own tailored plan. Nothing should be “cook-book” in this process. The ultimate aim should be to restore the patient to an active and pain-free lifestyle, whether this is in the context of a slow leisurely walk or full athletic activity (Davies, 2014a).

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4

Communicating the Treatment Plan

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Introduction

Veterinary patients with conditions that necessitate physical rehabilitation programs are often prescribed prolonged therapy. Sessions may be required daily or several times per week and treatment often extends over many weeks or months. Whether the patient needs to be brought into the rehabilitation practice for therapy or is prescribed a home exercise program, it requires a serious commitment from the patient's owner/caregiver. Rehabilitation does not guarantee a complete return to function, and setbacks and/or failures are not uncommon. Thorough owner/caregiver communication regarding the treatment plan, benchmarks of progress, and expected results is essential in

order to achieve a satisfactory outcome. The client/caregiver will almost always be asked to actively participate in some portion of the rehabilitation program. It is essential that the owner/caregiver has a good understanding of the basic theoretical, technical, and behavioral components of the exercises/techniques in order to safely, properly, and effectively assist with the rehabilitation plan. By understanding the basic rationale for the components of the plan, the client will have a greater investment and motivation to adhere to the prescribed plan. Incomplete or inaccurate understanding of techniques can lead to injury or failure to achieve a maximal recovery. Some owner/caregivers do not fully understand animal behavior and may misinterpret signs of stress in their pet. For the

plan to be effective and fun for both the owner/caregiver and the pet, minimizing behavioral stress is critical. It is often the veterinary technician's role to provide this important client education.

Diagnosis and Progress Assessments

Although clients do not always need or want a thorough understanding of the pathophysiology of the patient's disorder, at least a basic understanding of the diagnosis and dysfunction is important in order for the caregiver to be an active participant in the rehabilitation program. The veterinary technician should have a good understanding of the patient's diagnosis and the expected progression of healing (from specific tissue pathology to whole body function) in order to answer questions that arise during rehabilitation visits.

Progress assessment markers (Box 4.1), such as goniometry, pain scoring systems, muscle girth measurements, functional scores, and activities of daily living (ADL), are often used to document a patient's response to

therapy. These assessments are an excellent way to document and demonstrate progress, especially when patient improvement may be slow or subtle. Video recording the patient initially and periodically throughout the rehabilitation process can also aid understanding of improvements. Client input is often necessary when acquiring some of the subjective assessments, such as with a pain scoring system, ADL, and quality of life evaluation. The client should be included in the assessment and re-assessment process, keeping them updated on how their pet is responding through these markers. This helps to keep the client engaged in the pet's therapy program.

Goals

Every rehabilitation program should establish a set of goals for the patient. These goals, set by the rehabilitation veterinarian, are developed in conjunction with the client's expectations for the patient's return to some level of function. These goals are customized for each patient, and in our experience are based upon a current understanding of the pathophysiology of the patient's disorder, estimated tissue healing abilities, patient's ADL, and the owner/caregiver's expectations. The goals should be communicated clearly to the owner/caregiver to ensure owner/caregiver perception of program success. If the client and therapist have different goals in mind, they may not be satisfied with the outcome.

Goals must be reasonable and attainable. Unrealistic goals can cause clients to become discouraged and lead to abandonment of the rehabilitation effort. Setting several small goals over short periods of time, with re-assessments to confirm goals met, along with praise for reaching these small goals, can serve to keep clients motivated to continue the rehabilitation effort.

The Therapy Prescription

The discussion of the rehabilitation plan should be included in all phases of veterinary treatment planning, regardless of whether the

Box 4.1 Progress assessments

Subjective

- Observations: such as sit symmetry
- Pain assessment score
- Lameness scores
- Gait patterns
- Quality of life
- Activities of daily living (ADL)
- Body condition score
- Ease of position transitions (sit-to-stand)

Objective

- Girthometry
- Goniometry
- Force plate analysis
- Functional scales, such as Texas Spinal Cord Injury Scale or Proposed Stifle Function Scale
- Time/distance to fatigue
- Presence of tibial thrust/cranial drawer
- Presence of conscious proprioception

patient has a scheduled intervention, such as surgery, or is suffering from acute trauma or injury. The earlier the client is exposed to the idea and impact of rehabilitation, the more likely they are to become invested in the plan. Rehabilitation plans can vary in time intensity from one session per week to several sessions per day. In many cases, the plan requires a large time commitment from the individual performing the therapies. Although elaborate, time-intensive therapy plans can speed recovery for the patient, it is essential to take into consideration the client's time availability and lifestyle as well as the financial commitment when developing the plan. In one study of human physical therapy patients, one of the strongest factors of non-compliance was perceived barriers by the patient (Sluijs *et al.*, 1993). These perceived barriers were often related to the time involved in completing the exercises. Clients are often busy with work, family, and other activities. If clients are asked to commit more time than they have available, there is a risk that the client will either not perform all of the therapies, or become discouraged and abandon the rehabilitation effort entirely. If the client only performs a portion of the plan and neglects other segments, the program may become unbalanced, which may risk injury or failure to achieve a maximal recovery.

Many veterinary physical rehabilitation programs employ a combination of several different therapies, exercises, and modalities. Each one of these therapies offers the patient benefits, often in different ways. The veterinary technician should be well versed in the benefits and risks, if any, of each therapy so that this information can be fully communicated with the client. Clients often have questions about different modalities, especially when they may increase the cost or result in any temporary discomfort to the patient. Clients who do not understand the benefit of a particular modality are more likely to exclude or decline these treatments even if their pet may have benefited

Activity Restrictions

In rehabilitation practice, we often spend a large portion of time informing clients of

the importance of exercise in their pet's therapy. For example, we may stress the importance of strengthening core musculature to support orthopedic incompetency, and that regular exercise is important in promoting joint health in arthritic patients. Although we need to communicate the importance of exercise, we cannot overlook the critical importance of activity restriction. In most cases, restriction does not mean a complete lack of activity, it means a focus on a narrow range of acceptable activities. In fact, in most situations a balance of both exercise and activity restrictions are required for a successful outcome. It can be challenging to communicate this need for balance with clients; there may be a feeling of depriving their pet of enjoyment if freedoms are limited. If the importance of therapeutic exercise is overemphasized, the client may push the patient too hard, resulting in injury. If activity restrictions are overemphasized, clients can become fearful of doing too much, and the pet may not improve at an acceptable rate. The trained veterinary technician can play a valuable role in educating the client on the benefits and risks of both exercise and activity restriction.

Assistive Devices

As technology and interest in animal rehabilitation increases, so does the availability of assistive devices for animals such as orthotics, prosthetics, carts, and other mobility devices. Although the specific use and function of these devices is beyond the scope of this chapter and is covered elsewhere (see Chapter 10), it is important for the veterinary technician to be educated on their proper use and potential adverse effects. The veterinary rehabilitation technician will often be asked to demonstrate how to properly apply and use the devices, educate the pet owner on the appropriate duration of wear of the device, and to inform the client on how to monitor for discomfort and contact sores from these devices. Pictures and videos can help to communicate how to use and to jog the memory later.



Figure 4.1 Photo of a patient showing signs of stress during a therapy session (moving away from the therapist, while ignoring a food reward). The therapist is utilizing calming body language of sitting to the side, averting the gaze, and maintaining gentle contact with the patient. *Source:* Photo by Tracy Darling.

Keeping it Positive

Physical rehabilitation plans often entail exercises or procedures that are performed repeatedly, or as a series of repetitions over several sessions. In addition, many of the movements or positions that we ask of our patients are not natural, and may even be uncomfortable for the animal patient. Improper or uneducated application of these techniques or failure to recognize the subtle body language associated with increasing stress (Figure 4.1) can easily result in the patient developing an aversion to therapy.

This aversion can often be avoided by being alert to signs of stress and using reward-based training and low stress handling techniques (see Chapter 9) to keep patients encouraged and engaged in therapy. Animals can often be trained to willingly perform exercises with very little assistance. When pets stay engaged and perform tasks willingly, it becomes a much more pleasant experience for the client, patient, and rehabilitation professional. When demonstrating rehabilitation techniques with the client, the veterinary technician can show the client how to use rewards (e.g., food, toy, or praise) to motivate the patient (Figure 4.2). Another advantage of using training techniques to accomplish the techniques is that it provides mental exercise, which can be an effective means of releasing

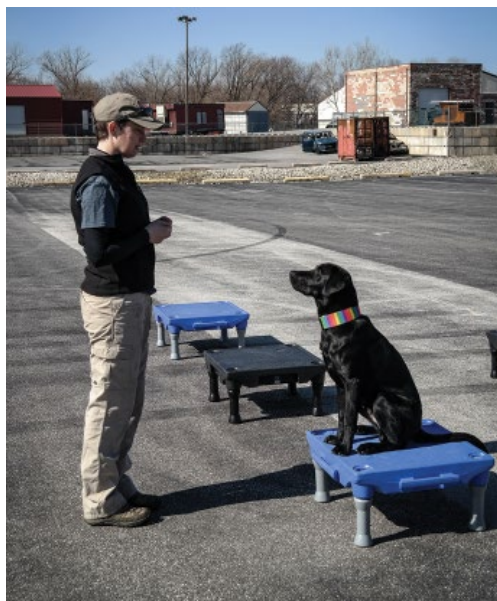


Figure 4.2 Reward-based training as part of a fitness and conditioning program.

energy in a patient that has activity restriction. Clients should also be educated about the early signs of fatigue or aversion to therapy so sessions can be abbreviated if needed. When used successfully, most patients will show up for therapy very excited to begin each session. This engagement of the patient in turn motivates the client, and therapy is more likely to continue and be successful.

Prescriptions and Supplements

Medications and supplements are often prescribed for veterinary physical rehabilitation patients to treat infections, reduce pain and inflammation, promote joint health, and reduce anxiety, as well as many other indications. The veterinary technician is an integral part of communicating the administration and proper usage of these important pharmaceuticals. Clients often have questions or concerns about why certain drugs or supplements are prescribed, how they will help their pet, whether there are drug interactions, what side-effects to look for, and how to administer medications. The veterinary technician can help increase compliance and treatment success when they have a thorough understanding of the commonly utilized medications and supplements. Sometimes medications are prescribed to be given “as needed.” Clients often need specific criteria to indicate to them when these medications should be administered. For example, a dog that is being treated with a non-steroidal anti-inflammatory drug (NSAID) to be given as needed, may become reluctant to rise or be slower climbing stairs after a therapeutic exercise session and then would benefit from receiving a dose of the NSAID. Discussions about the clinical signs of conditions such as discomfort or nausea can help to reassure clients that they have the knowledge to properly use these medications. Reinforcing these discussions by providing written instructions and/or handouts can ensure that medications are administered properly.

Adverse Events and Setbacks

When communicating with clients about physical rehabilitation treatment plans, a discussion about the possibility of setbacks or adverse events should always be included. Although our goals are always to improve a patient's condition, there may be times during therapy where clinical signs worsen, or a new injury occurs. Besides being detrimental for the patient, these incidents can be

devastating to clients and can strain the relationship between the client and the rehabilitation team. For these reasons, clients should always be educated about the potential for setbacks during therapy, what steps should be taken to mitigate them, how to identify signs of injury early, and how these setbacks will be addressed by the rehabilitation team. This information may be especially important when clients have elected physical rehabilitation in lieu of surgical options. Clients should be advised to discontinue activities at home and contact the therapist immediately upon the suspicion of an injury or adverse event with their pet.

Compliance

The nature of the condition may impact owner commitment and compliance. Any treatment including rehabilitation may be considered to be an optional extra (“icing on the cake”), or even an extreme measure because of the option of euthanasia in veterinary medicine. The presence of sudden disease versus gradual decline, the age of the pet, the general health of the pet, the relationship with the owner as well as the outlook for return of function may influence the initial commitment to rehabilitation of a pet. Once a client commits to a course of physical rehabilitation, the next challenge is to ensure compliance with the recommended plan.

In one study of human physical therapy patients, only 35% of adult patients fully complied with their prescribed regimen (Sluijs *et al.*, 1993). Compliance becomes even more complicated when the client providing the rehabilitation is not the patient, but rather the caretaker. This model of care is most similar to parents undertaking rehabilitation therapy for their children. Adherence to a recommended treatment plan has been associated with parental expectations and belief in the impact of the program. These factors are strongly influenced by the professionalism of, communication by, and trust in the health care professional (Dimatteo, 2004; Rabino *et al.*, 2013). It has also been proposed

that parents are most likely to adhere to their child's recommended physiotherapy if they perceive a severe disease threat (Rabino *et al.*, 2013); although not documented, in veterinary medicine, adherence may be impacted by the potential for return to acceptable function and quality of life. Based on human data, compliance is also influenced by how the therapist fulfills the individual's preference for an active versus passive role in health care of their dependent. A report published by the American Animal Hospital Association stated "using multiple approaches to client education and communication increases compliance" (Albers and Hardesty, 2009a, p. 16).

In both adults and children with chronic conditions, treatment adherence is less than 50% (Simons *et al.*, 2010). In older adults, the most common barriers to treatment adherence are time, transportation, perception of treatment efficacy, and fear of pain or injury (Austrian *et al.*, 2005). With children and adolescents suffering from chronic pain, the most common barrier was a negative attitude about the treatment recommendation; however early success with exercise increased the future compliance for physical therapy (Simons *et al.*, 2010). These findings emphasize both communication and design of a plan that can be implemented early and have measurable success. Based on the adage that "past behavior is the best predictor of future behavior," clients who have participated in fitness and conditioning programs with their dogs prior to an injury or illness are more likely to adhere to therapist recommendations.

Communication

There are several aspects of the rehabilitation program that are essential to communicate in order to maximize patient success. Shortcomings in any of these areas may compromise patient recovery. Setbacks during a therapy program are not uncommon, and often clients require frequent coaching and encouragement to keep them motivated. The

veterinary technician works very closely with the client and patient throughout the rehabilitation process, and is therefore a vital part of the communication effort.

Communication skills are at the heart of the effective home rehabilitation plan. All communication should be based on recognition of the value of the client, attention to the client's needs and concerns, acknowledgement of the client's message, empathy with the client, and a goal of providing the best care for both the client and the pet (Dreeben, 2011) (see also Chapter 7).

The communication style of the veterinary technician needs to be adjusted to each client. Clients will respond differently to verbal and non-verbal communication. Clients will be particularly attuned to the non-verbal communication between the technician and the pet, paying attention to the veterinary technician's level of comfort, empathy, and compassion. Any discrepancy between verbal expression of empathy and non-verbal cues will undermine the trust of the client and their confidence in the treatment plan. Although authentic expression of empathy (being able to understand and share the feelings held by the client) is one of the most powerful components of effective communication, the skilled veterinary technician also needs to be sensitive to cultural diversity and beliefs of the client. Verbal communication will be the foundation of sharing information and teaching the client. Effective verbal communication includes written and oral communication. The learning style and emotional state of the client will help determine how the client will most effectively receive information.

There are several recognized learning styles (Box 4.2). For people who learn by the hearing (aural) the therapist will need to articulate in words the full therapy plan, whereas verbal learners or patients under emotional duress may benefit from a combination of verbal and written communication and may be more likely to return with the need to "talk through the plan." Providing a conduit for additional follow-up communication

Box 4.2 Standard learning styles

Descriptions of learning modalities

- Visual: Learn by seeing shapes, pictures, three-dimensional objects
- Tactile (kinesthetic): Learn by touch, manipulating objects or the body, gestures
- Auditory: Learn by hearing words, sounds, tones, rhythms
- Verbal: Learn by articulating, reading or writing words
- Logic: Learn by deriving information from reasoning

Learning styles

- Accommodator=Concrete experience + active experiment: strong in “hands-on” practical doing (e.g., physical therapists)

- Converger=Abstract conceptualization + active experiment: strong in practical “hands-on” application of theories (e.g., engineers)
- Diverger=Concrete experience + reflective observation: strong in imaginative ability and discussion (e.g., social workers)
- Assimilator=Abstract conceptualization + reflective observation: strong in inductive reasoning and creation of theories (e.g., philosophers)

For further information see Barbe *et al.* (1979), Leite *et al.* (2010), and Kolb (2015).

Source: Adapted from https://en.wikipedia.org/wiki/Learning_styles and www.learning-styles-online.com.

such as an email address is extremely beneficial. When explaining and demonstrating therapeutic exercises, some clients may be able to learn from watching the therapist perform the exercise, whereas others need to have the hands-on experience. Demonstration can be effective, especially if the therapist describes each action and demonstrates common errors to avoid. Many individuals, however, will require the kinesthetic experience of having their hands/body move in the proper motion. This can be augmented by the therapist guiding the movements, or by feedback for the caretaker of watching themselves in a mirror or on video. The visual reminders of videos, photos, or drawings are particularly important to visual learners and provide a guide to remind all clients of the proper technique that they have learned. Some clients will learn best by understanding the theory behind the actions; for example, if it is explained how exercises that encourage eccentric muscle contraction, such as crouching through a tunnel, are used to strengthen muscles of the limbs and core (Figure 4.3).

In a study of children with cystic fibrosis, parents found written and video instructions

were a valuable adjunct to verbal and practical training; however, the documents must be professional, current, and credible (Tipping *et al.*, 2010). Information overload is a risk, especially for a client who is dealing with a sudden and unexpected event. Repetition and patience will be critical in the communication process.

Note: it is important to emphasize the privacy of clinic-provided videos; they should not be shared online or used for therapy on other animals.

Continued advances in technology give us the ability to communicate with clients in a variety of ways. Utilizing multiple communication modalities can help enhance client understanding and retention of what can sometimes be an overwhelming amount of information. There is so much information about clinical findings, diagnosis, treatment options, side-effects, and many other aspects of a patient's condition that cannot always be communicated effectively using one form of communication alone. One of the most common ways we communicate with clients is verbally, either over the telephone, or in person during an appointment session.



Figure 4.3 A patient performing a crawl exercise for a treat reward. *Source:* Photo by Tracy Darling.

The use of verbal communication is an effective way to relay information, and it is also a means to build relationships with clients. Pairing verbal communication with something visual, such as anatomical models or pictures, can help clients' understanding of disorders and disease processes. When communicating a treatment plan with a client, hands-on instruction in how to properly perform exercises and other therapies are essential in developing client

confidence in carrying out such treatments at home (Figure 4.4).

Online videos demonstrating commonly used techniques are a great way to reinforce hands-on instruction, as they can be accessed on-demand by the client after they leave the clinic. Videos can be customized by the rehabilitation clinic, or there are many great online resources of topics related to physical rehabilitation that can be shared with clients.



Figure 4.4 Veterinary technician providing hands-on instruction in the execution of assisted exercise.

Treatment schedules can often be quite elaborate. It is helpful to provide a treatment schedule in written format. Home exercise plans can be printed and handed to the client or sent via email after each visit.

The nature of the condition and expectation for the necessary duration of home rehabilitation is variable. Lessons from parents of children with cystic fibrosis provide some insights into approaches to client education with severely affected patients or those with chronic disease (Tipping *et al.*, 2010). In a qualitative study evaluating factors that impair delivery and retention of physiotherapy education of parents and impact effective home physiotherapy treatment, three major themes were identified: (i) transition from one life or disease stage or even location to another, (ii) psychological distress often associated with the learning process and the underlying condition, and (iii) social connectedness with the health professional and social networks. In the case of pets, no doubt the connectedness with the pet is also a major factor.

Discussing Finances

When discussing treatment plans with clients, it is often necessary for the veterinary technician to discuss the estimated cost of services. Finances can be a stressful topic for some clients. In our experience, there are some ways to approach the topic without added difficulty. Focusing on the needs of the patient and explaining how the services will benefit the patient is an excellent way to increase client acceptance (Albers and Hardesty, 2009b, p. 19). Clients may be more willing to accept the cost associated with a treatment when they can appreciate the value of the service. For example, a therapy visit of US\$150 might sound expensive to an client, especially when told that their pet will need weekly visits for 12 weeks. On the other hand, if they first learn that each 1-hour therapy visit will include massage therapy, therapeutic laser, an underwater treadmill session, and assisted exercise, they may be surprised

to hear that each visit will only cost US\$150. Disclosing all costs and obtaining written consent before providing services are good ways to avoid financial conflict after therapy sessions are completed.

Follow-Up

In a study of rehabilitation compliance in human athletes, patients with emotional support were more likely to adhere to their rehabilitation regimen (Byerly *et al.*, 1994). This is likely to be also the case with caregivers of veterinary patients undergoing rehabilitation. Client communication should not end when the client leaves the clinic. The veterinary technician can be utilized to maintain an open line of communication with clients in between rehabilitation visits. Clients often leave an appointment excited to start rehabilitating their pet, but may be overwhelmed or nervous about carrying out treatments properly. Clients also may have questions after they leave, but might not be inclined to reach out until their next visit. A follow-up phone call or email the day after an initial visit to check-in will not only help answer client questions, but will keep the line of communication open and reassure the client that you are behind them 100%. This support should continue throughout the treatment program. The veterinary technician can provide valuable support during follow-up therapy visits. Technicians often perform many of the functional assessments and can discuss these results with the client, providing praise for improvements and being prepared to discuss the potential causes for lack of improvements.

Conclusion

The way in which a treatment plan is presented and communicated with clients will have a powerful effect on patient outcome and should not be overlooked. There are many ways that the rehabilitation team can

use effective communication to maximize success in the veterinary rehabilitation patient. This aspect of rehabilitation practice is often tasked to the veterinary technician.

The unique knowledge and perspective of the trained veterinary technician makes them well suited for this important responsibility.

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5

Manual Therapy Treatment

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Introduction

Canine rehabilitation treatment can be categorized into three groups: therapeutic exercise, modalities, and manual therapy. This purpose of this chapter is to discuss manual techniques commonly used in rehabilitation. As a treatment, manual therapy utilizes skilled hands-on techniques. Safe and effective use of these techniques requires specific training and an understanding of indications, contraindications, expected results, and red flags.

tion of pressure, force, and movement applied to soft tissues and joints. These techniques include soft tissue mobilization (STM), passive range of motion (PROM), muscle stretching, and joint mobilization/manipulation. Joint mobilization and joint manipulation will be covered briefly in this text as they can be performed only by an appropriately trained veterinarian (trained in rehabilitation or manipulative therapies), a physical therapist with training in animal rehabilitation, or a chiropractor trained in animal chiropractic.

What is Manual Therapy?

Manual therapy is used to address musculoskeletal pain and disability. Treatments include a variety of techniques involving a combina-

Goals of Manual Therapy

Before initiating any treatment, the therapist must specify the goal of the treatment. The specific goals of each technique will be

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discussed later in this chapter. Typically, one or more of the following goals are expected with manual therapy:

- Increased circulation
- Decreased pain
- Decreased swelling
- Increased range of motion (ROM)
- Increased flexibility
- Improved proprioception
- Relaxation and decreased anxiety.

Expected Results

Once a treatment is completed, the therapist must determine if the desired effects were achieved. Unlike other types of treatment (e.g., therapeutic exercise with a goal of strengthening), change is expected to occur relatively quickly, within one to two manual therapy sessions. Using the “test–re-test” model of data collection will help to determine the effect of the treatment (Hendrickson *et al.*, 1993). This technique is applied by performing an objective measurement prior to the treatment (pre-test) and again after the treatment (re-test). In so doing, one can objectively determine if the treatment goals were accomplished.

For example, if the treatment goal is to increase shoulder extension PROM, PROM is measured with a goniometer prior to the treatment and again afterwards. Improved ROM should be noted.

If improvement is not noted, the treatment and re-test measurement should be repeated. If there is still no improvement, your supervising therapist should be notified, as the joint may need further evaluation to properly identify the limiting structure and institute a change in the treatment plan.

Preparing the Patient for Manual Therapy

Building trust with the patient is critical to a successful treatment. Manual work requires that the patient be willing to be touched. Some patients are uncertain or anxious in a new environment. As veterinary technicians

and nurses, you are acutely aware that animals cannot have the treatment explained to them and need other forms of reassurance. Anxiety may be magnified when someone other than their owner attempts to handle them. Thus, the following suggestions are recommended for creating a non-threatening environment for the patient:

- Allow the patient to explore their surroundings before being handled.
- Allow the patient to get used to your voice and presence before touching them.
- Use a quiet treatment space and speak in a soft, reassuring voice.
- Encourage the patient to settle in the treatment area using owner, treat, and toy motivation; do not use force (Yin, 2009).
- Allow the patient to move a little; apply minimal restraint.
- Start with soft, quiet hands; move your body and hands slowly.
- Work on non-painful areas first.

See Chapter 9 for more information.

Manual Techniques

Soft Tissue Mobilization

STM is a therapeutic technique in which superficial and deep layers of soft tissues are manually manipulated. The goal of treatment is to normalize tissue function. The tissues addressed include muscle, tendon, ligament, and fascia. STM is often grouped under the umbrella term “massage;” however, massage is the general concept, whereas STM is the specific how and what. It is used within a total systematic therapeutic approach, designed to achieve functional goals. These techniques are used in conjunction with additional therapeutic techniques as part of a complete treatment plan (Hesbach, 2014).

Effects of Soft Tissue Mobilization

STM is purported to affect several of the body’s systems and have a range of mechanical, physiological, and psychological benefits. Authors describe effects on the muscular, nervous, lymphatic, circulatory, and endocrine systems (Fritz and Grosenbach, 2009; Salvo, 2012). By affecting these systems,

mechanical, physiological, and psychological results are noted. Much of the information regarding STM is based on hypothesis and anecdotal experience. Underlying tissue changes have not been well researched to date. Short-term effects are demonstrated subjectively in human medicine; however, there are no studies to support the efficacy of long-term effects. There is a tremendous need for further research in STM.

The proposed physiologic effects of STM can be categorized into three groups: reflexive (indirect), mechanical (direct), or both (Fritz, 2004). Reflexive responses occur as a result of stimulating the nervous or endocrine systems. Mechanical effects are produced in response to direct force to the body. The various types of mechanical forces are discussed below. Mechanical effects can be measured objectively (Fritz, 2004).

Authors refer to a variety of responses, including (DeLisa *et al.*, 2005):

- vascular changes,
- decreased edema,
- breakdown and prevention of adhesions,
- decreased pain, due to the release of endorphins,
- stimulation of cutaneous mechanoreceptors and muscle spindle receptors in superficial skeletal muscle,
- reflexive vasodilation, and
- patient relaxation.

Current research suggests that the effects of STM include decreased motor unit activity of the muscle, increased tissue extensibility, and increased ROM (Sefton *et al.*, 2011).

Studies have demonstrated a reduction of pain by increasing T lymphocyte proliferation, both in diseased and in healthy patients (Austin *et al.*, 2012). It is speculated that this effect in massage is related to stress reduction or lymphatic activation, but the underlying mechanisms have yet to be verified (Billhult *et al.*, 2009; Wang *et al.*, 2014).

Compressive STM techniques have been shown to increase immunomodulatory effects on rat skeletal muscle, offering an alternative to anti-inflammatory medication (Waters-Banker *et al.*, 2014).

Types of Forces

STM techniques are designed to deliver one or more kinds of force with the intent of creating specific physiological effects (Fritz and Grosenbach, 2009; Simancek, 2013). There are myriad STM techniques, each designed with a specific purpose in mind. They are differentiated by the type of force being applied, these forces consisting of compression, tensile, torsion, shear, and bend.

- *Compression* is a force that compresses soft tissues together, referred to as tissue approximation. Tissue approximation is used to support circulation, stimulate nerve function, and restore connective tissue mobility (Fritz and Grosenbach, 2009). Compression is used in several techniques, such as effleurage, petrissage, and ischemic compression (Simancek, 2013).
- *Tensile* is a force that occurs when two ends of a structure are pulled in opposite directions. It is a lengthening force. Therapeutically, this force is used to promote fiber realignment and extensibility (Simancek, 2013).
- *Torsion* is a force that involves twisting or turning of soft tissues. This force is used in kneading techniques (Simancek, 2013).
- *Shear* is a sliding force in which two structures slide across each other, creating friction. This force is used in friction massage (Simancek, 2013).
- *Bending* is a force that combines compression and tension such that a compressive force is applied to one side of the structure while a tensile force is applied to the opposing side. These forces are applied perpendicularly across the tissue (Simancek, 2013).

Contraindications

If the appropriate amount of pressure is applied during STM, there are very few contraindications (Furlan *et al.*, 2002). A few that should be considered include:

- acute inflammation,
- skin or deeper tissue infections,
- burns, and
- deep vein thromboembolic disease.

Techniques

Passive Touch Passive touch allows the person performing massage and the patient to make an initial connection. The patient is allowed to settle, become accustomed to the therapist's hands and respond to the calm, quiet environment. The therapist relaxes and hopefully this is communicated to the patient. Allow both hands to rest on the patient. If the patient becomes anxious, spend a few minutes gently stroking the patient to allow them to calm down (Robinson and Sheets, 2015).

Effleurage Effleurage is a French word meaning “to skim” or “to touch lightly on.” It is a gliding stroke characterized by rhythmic, long, broad, continuous strokes of light to moderate pressure. Effleurage is often performed parallel to the fibers of the muscle and on larger muscles groups or throughout an entire limb. The therapist's hands are open and relaxed, working in a hand-over-hand pattern (Salvo, 2007; Fritz and Grosenbach, 2008) (Figure 5.1).

Pressure and direction can be varied, making this technique quite versatile (Fritz, 2004). A more superficial effleurage stroke is presumed to stimulate cutaneous mechanoreceptor firing. Increased mechanoreceptor nerve firing into the spinal cord may reduce central nervous system input of smaller diameter, slower signaling pain fibers, thus reducing the perception of pain (Gay *et al.*,

2015). Because a deeper stroke can mobilize and aid resorption of extracellular fluids, deep stroking is recommended only in the direction of venous or lymphatic flow, whereas superficial strokes can be in any direction (DeLisa *et al.*, 2005). Effleurage can be used as an introductory technique when initiating a treatment, as a method of directing fluids toward the heart or lymph system, or as a means of promoting relaxation. It allows the therapist's hands to palpate the body and to explore underlying tissues.

The postulated effects of effleurage (Fehrs, 2010a) are that it:

- moves blood and lymph toward a desired area and removes metabolites;
- reduces pain by decreasing edema and therefore nociceptor firing and releases endorphins;
- promotes relaxation via the reflex effects that occur with stimulation of nerve endings, including the release of neurotransmitters and local hormones (e.g., vasopressin and oxytocin); and
- inhibits muscle spasm via prolonged pressure of the Golgi tendon organ and muscle spindles (Chaitow, 2010).

Petrissage Petrissage is a French word meaning “to knead.” It represents a classification of strokes that include a variety of movements



Figure 5.1 Effleurage.

Figure 5.2 Petrissage.

such as lifting, kneading, compressing, wringing, and rolling of soft tissues. Unlike effleurage, petrissage is a compressive, stimulating stroke.

Petrissage techniques such as lifting, kneading and skin rolling are performed by “picking up” the soft tissues between the thumb and the flat side of the index finger(s) or between the two hands of the therapist (DeLisa *et al.*, 2005) (Figure 5.2).

The postulated mechanical effects of petrissage include:

- increased tissue extensibility,
- increased circulation,
- muscle relaxation,
- decreased pain, and
- improved cellular nutrition (Salvo, 2007).

Tapotement Tapotement is based on the French word “Tapoter,” meaning to tap or pat. This technique involves rapid, repetitive striking of the tissues and is used as a stimulating technique. There are several variations of this technique used in human medicine, such as tapping, hacking, slapping, cupping, pinching, or plucking. Tapping is the gentlest technique and is used commonly in canine rehabilitation to stimulate a muscle in orthopedic or neurologic cases.

In this technique, the tips of fingers are used to tap the muscle at a brisk rate (Fritz, 2004).

The postulated mechanical effects of tapotement (Fehrs, 2010b; Salvo, 2007) are that it:

- increases circulation,
- warms and softens underlying tissue,
- stimulates nerve endings, and
- produces tiny muscular contractions that result in increased muscle tone. This is thought to happen via stimulation of mechanoreceptors in the fascia and Golgi tendon organs.

Friction Massage Friction massage was first described by Hippocrates, but was popularized by Dr. James Cyriax, a British orthopedic physician who worked with physical therapists to develop a number of orthopedic evaluation and treatment techniques including friction massage. This technique is commonly used to maintain/improve mobility within ligaments, tendons, and muscles and to prevent adherent scars from forming (Chamberlin, 1982; Robinson and Sheets, 2015). Instrument-assisted cross friction massage has been shown to increase tissue perfusion in collateral ligaments of the rat knee (Loghmani and Warden, 2013). In addition, friction massage aims to realign fibers caused by micro tearing of injured tissues. Friction massage is usually performed as a deep technique affecting multiple layers of tissue. The primary effect is the application

of shear stress to underlying tissue, particularly at the interface between two types of tissue (DeLisa *et al.*, 2005).

The tip or “ball” of the finger or thumb is used in this technique. Enough pressure is applied to the specific tissue such that the shear force is directed to deeper, rather than superficial, layers of the tissue. The finger does not glide across the tissue but rather creates shear force transmitted to a deeper interface. The therapist’s fingers and the patient’s skin move as one (Chamberlin, 1982). The direction of force can be parallel to, perpendicular to, or diagonally across the fibers. If the treatment goal is to realign fibers (tendons, ligaments, or muscles), cross friction massage is specifically performed perpendicular to the tissue fibers. In human medicine friction massage lasts approximately 5–15 minutes (Goats, 1994). This may not be realistic for animal patients.

The postulated mechanical effects (Salvo, 2007; Chamberlin, 1982) of deep massage include:

- fiber realignment,
- maintenance or increase in mobility,
- break down of scar tissue, and
- hyperemia.

Ischemic Compression Ischemia is a restriction of blood supply to the tissue. Ischemic compression is performed on a tender point or trigger point. This technique intentionally blocks the blood supply to the painful nodule such that upon release there is a resurgence of blood (Travell and Simons, 1983).

The pad of the thumb or finger is used to apply direct, static, moderate-to-heavy pressure over the tender point or trigger point. Pressure is gradually increased according to patient tolerance and continued for up to 1 minute. Following the treatment, the muscle should be gently stretched so the muscle can “remember” its new length (Travell and Simons, 1983).

The postulated mechanical effects of ischemic compression are that it:

- increases local circulation,
- removes waste products,

- increases oxygen, and
- promotes healing (Travell and Simons, 1983).

Myofascial Trigger Points

Trigger point release (TPR) techniques were popularized by the physician Janet Travell as early as 1942 because of her work on myofascial pain. She defines a trigger point as a hyperirritable point within a taut band of a muscle (Travell and Simons, 1983). Trigger points are classified as “active” or “latent.” An active trigger point is painful and has a predictable pattern of referred pain, which “frequently occurs within the same dermatome, myotome or sclerotome as that of the [trigger point] but does not include the entire segment” (Travell and Simons, 1983). An active trigger point causes a clinical pain complaint. It prevents full lengthening of the muscle, weakens the muscle, and mediates a local twitch response of muscle fibers when adequately stimulated (Bron and Dommerholt, 2012). When compressed within the patient’s pain tolerance, it produces referred motor and often autonomic phenomena, generally in its pain reference zone. A latent trigger point is clinically quiescent with respect to spontaneous pain; it is painful only when palpated (Bron and Dommerholt, 2012). A latent trigger point may have all the other clinical characteristics of an active trigger point and always has a taut band that increases muscle tension and restricts range of motion. For further information see Chapter 25.

Joint Range of Motion

Joint ROM refers to the movement that occurs at a joint (flexion, extension, abduction, adduction, internal rotation, external rotation) (Norkin and Levangie, 1992).

Joint motion can be passive or active. Passive ROM (PROM) requires a second party (the assistant) to move the joint through the desired ROM *without* participation of the patient. No muscle contraction occurs with PROM. If the patient assists or resists the motion, it is no longer “passive.” Active ROM (AROM), on the other hand, requires the

patient to contract muscles to move the joint through its available range.

Joint ROM consists of both osteokinematic and arthrokinematic motion. Osteokinematic motion occurs when bony segments move about a joint axis. Arthrokinematic motion is the movement that occurs between opposing joint surfaces *within* the joint. Arthrokinematic motion is passive and cannot be visualized (see section on Joint Mobilization).

Importance of Range of Motion

Normal joint ROM is essential for normal osteokinematic movement. Normal joint motion is also necessary for proper function of associated tissues (muscle/tendon, ligament, disc, meniscus, cartilage, etc.). If joint ROM is limited, abnormal stress can occur at the joint and/or related soft tissues. The patient likely attempts to make up for lost motion by increasing movement within the joint or at an adjacent joint. This is referred to as compensation. Compensation can result in secondary joint hypermobility, joint irritability, abnormal use or stretch of related muscles, and pain (Weiner, 2001; DiBerardino *et al.*, 2012; Wise, 2015). Early attention to limited ROM is important to avoid the secondary effects.

What can Limit Joint Range of Motion?

ROM can be restricted by several structures and conditions. These include swelling, muscle/tendon, ligament, capsular tightness, pain, meniscus, and a joint mouse. It is important to identify the restricting structure so that treatment can be guided accordingly. The source of limitation will be determined by assessing end-feel at the end range of osteokinematic motion (Sprague, 2013). Overpressure is applied into the restricted range and the sensation or “feel” of the restriction in the therapist’s hands is noted. The therapist identifies the limiting structure by “feel.” End feels are categorized as muscle/tendon, soft tissue approximation, capsular, muscle spasm, boggy, bony, springy, or empty (O’Sullivan and Schmitz, 1994). End feels transmit specific sensations to the therapist’s hands depending on the source of a joint’s stiffness at the end ROM (Sprague, 2013).

Passive Range of Motion

Maintaining or improving joint ROM is important in preventing the negative effects of immobilization. PROM is used as a manual treatment technique with the aim of maintaining or regaining joint ROM. Because this technique is directed at the joint, the joint must be positioned such that active muscular restriction is minimized. For example, when treating a restriction into elbow extension, the shoulder must be placed in a neutral position. If the shoulder is placed in a flexed position, elbow extension will be limited by muscle tightness of the biceps. PROM is performed within the patient’s comfort level so that muscle splinting (a protective response involving active muscle tightening) does not occur (Figure 5.3).

Treatment Technique The technique used when performing PROM entails:

- stabilizing the proximal bony segment;
- moving the distal bony segment to its end range;
- maintaining the proper plane of motion (not allowing abduction/adduction or rotation); and
- working within a pain-free range of motion (to avoid muscle splinting) (Reese and Brady 2010).

Manual contact or hand placement is critical when performing PROM. Keep in mind that the distal bone is moved on a stabilized proximal bone. Therefore, one hand will stabilize the proximal bony segment and the other hand will support the limb while moving the distal bony segment. Support of the limb usually occurs at the joint just distal to the one being treated. For example, in the case of shoulder extension, one hand will stabilize the scapula (acromion) while the other hand supports the limb at the elbow and moves the humerus into extension. If the scapula is not stabilized, motion will occur at both the scapulothoracic joint and the glenohumeral joint. When performing PROM, the therapist must be mindful of moving the limb in the proper plane of motion. In the example of shoulder extension



Figure 5.3 Passive range of motion.



Figure 5.4 Glenohumeral extension.

PROM, the humerus should remain in the sagittal plane, not allowing shoulder adduction, abduction, or rotation (Figure 5.4).

It is important to watch for any cues of patient discomfort. These may be as subtle and a change in respiratory rate or licking of the lips or as blatant as muscle splinting or vocalization. If the patient splints (actively contracts muscles), the movement is no longer passive.

Determining the Effectiveness of Treatment

The test and re-test concept (as explained in the section on Soft Tissue Mobilization) is

used to determine the success of a treatment with PROM. Objective changes in ROM are measured with a goniometer and documented in degrees. Goniometric measurements can be taken before and after treatment to determine the effectiveness of the treatment (Figure 5.5).

The term goniometry is derived from two Greek words: *gonia*, meaning angle and *metron*, meaning measure. A goniometer is an instrument used to measure angles. Within the field of human physical therapy, goniometry is used to measure the total amount of available motion at a specific joint (Dutton, 2015). Goniometers come in a variety of

Figure 5.5 Goniometer.**Figure 5.6** Stifle flexion measurement.

sizes and shapes and are usually constructed of either plastic or metal.

Measurement of PROM requires proper placement of the goniometer at the joint being measured. A goniometer has two arms that move around a fulcrum. A list of the bony landmarks is used to align the goniometer with a proximal bony landmark, the joint axis and a distal bony landmark. The proximal arm or “stable arm” is lined up with the proximal stabilized bony segment and does not move during measurement. The distal arm or “moving arm” is lined up with a distal bony segment and will move as the joint is taken through ROM. The axis or fulcrum is lined up with the

approximate joint axis. The therapist must maintain the proper PROM manual contacts discussed earlier as well as stabilize the goniometer on the limb (Figure 5.6).

ROM Standards

In human medicine, normal joint ROM has been well documented and specific values are available (Norkin and White, 2009). Normal joint ROM of the canine patient is not well established due to conformation variability by breed (Thomas *et al.*, 2006) (Table 5.1). Two canine studies have looked at the reliability of goniometry and have documented ROMs by joint for a particular breed in dogs without lameness or radiographic

Table 5.1 Commonly used goniometric bony landmarks.

Joint	Proximal arm	Axis	Distal arm
Shoulder	Spine of scapula	Acromion	Lateral epicondyle
Elbow	Greater tubercle	Lateral epicondyle	Lateral styloid
Carpus	Head of radius	Lateral styloid	5th metacarpal
Hip	Midline of ilium	Greater trochanter	Lateral epicondyle
Stifle	Greater trochanter	Lateral epicondyle	Lateral malleolus
Tarsus	Head of fibula	Lateral malleolus	5th metatarsal

Table 5.2 End-feel guidelines.

Joint	Motion	NL End-feel
Shoulder	Flexion	Soft tissue
Shoulder	Extension	Capsular
Elbow	Flexion	Soft tissue
Elbow	Extension	Bony
Carpus	Flexion	Capsular
Carpus	Hyperextension	Capsular
Hip	Flexion	Soft tissue
Hip	Extension	Capsular
Stifle	Flexion	Soft tissue
Stifle	Extension	Hard capsular
Tarsus	Flexion	Muscle–tendon
Tarsus	Extension	Capsular

abnormalities. These measurements were correlated with radiographic measurements (Jaeger *et al.*, 2002; Thomas *et al.*, 2006) (Tables 5.2 and 5.3). Due to conformational variability, between breeds we must rely on comparison to the contralateral side if that side is considered normal. Normal would be considered the absence of an injury or condition that might affect ROM. Goniometry has also been researched in cats and osteoarthritis correlates with reduced ROM (Jaeger *et al.*, 2007; Lascelles *et al.*, 2012).

Frequency, Intensity, and Duration of ROM

Treatment prescriptions must include specific parameters of frequency, intensity, and

duration (APTA, 2014). Frequency refers to how often ROM will be performed (i.e., daily, three times/day, every other day etc.). Intensity refers to the number of repetitions that will be performed. Duration refers to the number of sets of the prescribed repetitions.

For example, for shoulder extension PROM, 10 reps $\times$ 1 set, 3 $\times$ /day:

- Frequency 3 $\times$ /day
- Intensity 10 repetitions
- Duration 1 set (of 10 repetitions).

Precautions/Contraindications of PROM PROM is a relatively safe technique with few precautions or contraindications. Caution should be used in the case of acute tears, fractures, and surgery. In human medicine,

Table 5.3 Range of motion (degrees) of various appendicular joints measured by goniometry in 16 healthy Labrador Retrievers.

Joint	Position	Mean	95% Confidence interval of the mean	Median
Carpus	Flexion	32	31–34	32
	Extension	196	194–197	196
	Valgus	12	11–13	12
	Varus	7	6–8	7
Elbow	Flexion	36	34–38	36
	Extension	165	164–167	166
Shoulder	Flexion	57	54–59	57
	Extension	165	164–167	165
Tarsus	Flexion	39	37–40	38
	Extension	164	162–166	165
Stifle	Flexion	42	40–43	41
	Extension	162	160–164	162
Hip	Flexion	50	48–52	50
	Extension	162	160–164	162

Source: Adapted from Jaeger *et al.* (2002).

gentle, pain-free PROM is used in each of these instances to stave off adhesion, contracture formation, and sluggish circulation leading to a prolonged recovery time (Kisner and Colby, 2012).

Stretching

Stretching is a manual technique used to increase the extensibility of muscle fibers (Page, 2012). Whereas PROM affects the joint and its associated soft tissues (joint capsule and ligament), stretching primarily affects muscle and associated tendons. Decreased flexibility or muscle tightness can occur passively or actively. Passive tightness may be the result of postural adaptation or scarring and active tightness may result from overuse, injury, or spasm (Page, 2012). As a treatment, the goal of stretching is to increase the flexibility of the tight muscle.

Importance of Flexibility

As discussed in the previous section, ROM can be limited by several structures, one of which is muscle. If muscle tightness restricts ROM, compensation may occur, resulting in

abnormal stress to the joint and/or related tissues (Weiner, 2001; Jarvis *et al.*, 2013; Wise, 2015). Therefore, it is important to address issues of decreased flexibility early in the rehabilitation process.

Treatment Techniques

There are several techniques used to increase muscle flexibility, such as static stretching, modified stretching, and active stretching.

Static Stretching A static stretch involves holding a muscle in a stretched position for a prolonged period. Static stretching has been found to be effective in increasing muscle flexibility (Page, 2012). A word of caution has been offered regarding this type of stretching. Research has demonstrated that a static stretch can result in “stretch-induced strength loss” (McHugh and Cosgrave, 2012). This suggests that static stretching would be ill-advised prior to an athletic event. A dynamic stretch is recommended in this case.

A static stretch is performed by positioning the muscle in the stretched position and maintaining the position for a prolonged

Table 5.4 Thoracic limb muscle stretches.

Muscle group	Movement
Brachiocephalicus	Extend the head and neck, pull front leg forward
Rhomboids	Glide scapula ventrally
Biceps	Flex shoulder, extend elbow
Supraspinatus	Flex shoulder with slight internal rotation at humerus
Infraspinatus/teres minor	Extend shoulder, adduct limb
Deltoid (spinal head)	Extend shoulder, adduct limb
Triceps (long head)	Flex elbow then flex shoulder joint
Latissimus dorsi/teres major	Extend shoulder, slight external rotation and abduction at humerus
Subscapularis	Flex shoulder, externally rotate/slightly abduct humerus
Superficial pectoral	Adduct the forelimb, assist in protraction and retraction
Deep pectoral	Extend shoulder, abduct limb at humerus
Carpal flexors	Flex elbow, extend carpus
Carpal extensors	Extend elbow, flex carpus
Digital flexors	Flex elbow, extend carpus and digits
Digital extensors	Extend the digits

period. Research suggests that the effects of a 30-second sustained stretch are maintained significantly longer than those of a 15-second stretch. Stretching for longer than 30 seconds does not have increased benefit for human muscles (Bandy and Irion, 1994; Page, 2012). Canine muscles have different fiber type composition to humans and so may respond differently to stretch, but because current evidence is lacking we extrapolate information from another species. As therapists, we should use our palpation skills to assess the effects of stretching.

The position of stretch is determined by reversing the action(s) of the muscle (Tables 5.4 and 5.5). For example, the biceps brachii muscle has two proximal head attachments and extends from the distal end of the scapula to the proximal end of the radius. When it contracts, it flexes the elbow joint (Coville, 2016). This treatment technique must be comfortable for the patient so that splinting (contracting the stretched muscle) does not occur as this would negate the intent of the treatment (Figure 5.7).

Modified Stretching It is important to note that the safest and most effective stretch targets the belly of the muscle, rather than the tendon. Excessive loading of a tendon can result in irritation of the tendon (Sharma and Maffulli, 2005). The modified stretch assures that the stretch occurs in the belly of the muscle.

This technique is a two-step process. First, the muscle is placed in the stretch position until it is taut (not tight). In the case of a two-joint muscle, the proximal joint is positioned first, followed by the positioning of the distal joint (Figure 5.8). For example, the hamstring muscle group is stretched by first flexing the hip. Then the stifle is extended until the muscle feels taut. Once this position is achieved, the second step is to perform STM at the belly of the muscle with the intent of manually “bending” or stretching the muscle belly.

Active Stretching For the dog, cat (or horse) that would “rather do it themselves” a dynamic or active stretch can be taught. This

Table 5.5 Pelvic limb muscle stretches.

Muscle group	Movement
Iliopsoas	Extend hip, internally rotate femur
Semimembranosis	Flex hip, extend stifle. Slight abduction at femur
Semitendinosus	Flex hip, extend stifle. Slight external rotation at femur
Biceps femoris	Flex hip, extend stifle. Slight adduction and internal rotation at femur
Gracilis	Flex hip, extend stifle, abduct limb at femur
Sartorius (cranial band)	Flex stifle, extend hip
Pectineus/adductors	Abduct limb at femur
Gastrocnemius complex	Extend stifle, flex tarsus
Superficial digital flexor	Extend stifle, flex tarsus, extend digits
Tibialis anterior	Extend tarsus
Digital flexors	Flex the digits
Digital extensors	Extend the digits

Figure 5.7 Biceps stretch.



Figure 5.8 Hamstrings stretch.



type of stretching is recommended prior to an athletic event.

The targeted muscle is stretched by luring the patient into a position of stretch. For example, the hip flexor muscle group can be stretched by encouraging the dog to put his front legs up on a chair, bringing the hips into an extended position. Another example is a stretch of the triceps and latissimus dorsi muscles by luring the dog into a play bow position. The disadvantage of active stretching is that the origin of the muscle is not stabilized, making the stretch less specific to one muscle (Figures 5.9 and 5.10).

Frequency, Intensity, and Duration

There are varying opinions regarding frequency and duration of stretching for optimal results in gaining and maintaining flexibility. With regard to frequency, human research demonstrates that stretching more than once each day does not have significant benefits. In addition, stretching four times a week was shown to improve flexibility 82% greater than stretching only twice a week (Brady, 2013). An earlier study demonstrated that there was no increase in muscle length after 2–4 repetitions (Taylor *et al.*, 1990).



Figure 5.9 Psoas stretch.



Figure 5.10 Play bow.

Applying these findings to the canine patient, the recommended stretching prescription is:

- Frequency 4 times/week
- Intensity per patient comfort (to avoid splinting)
- Duration 30-second stretch.

Precautions/Contraindications

Like PROM, stretching is generally a safe treatment technique if performed within the patient's comfort level. The following precautions and contraindications should be considered.

Precautions

- Joint hypermobility
- Newly healed fractures
- Postoperative tendon repair
- Vigorous stretch post immobilization.

Contraindications

- Bony block
- Recent fracture
- Acute pain with movement
- Acute inflammation or infection
- Hematoma.

Joint Compression

Joint compression is a mobilization technique in which opposing surfaces of a joint are manually approximated (Kaltenborn, 1989). The goal of the treatment is to decrease pain, decrease swelling, improve proprioception, and simulate weight bearing.

This technique is performed by stabilizing one joint surface and mobilizing the opposing

surface such that the joint is compressed. No joint motion occurs. Joint compressions consist of quick oscillations or pulses and can be performed in non-weight-bearing (NWB) or weight-bearing (WB) positions. In the NWB position, the proximal bone is stabilized and the distal bone is mobilized through a long lever arm. Conversely, in the WB position the distal bone is stabilized (on the ground) and the proximal bone is mobilized. For example:

- Stifle joint compression in NWB position: The proximal hand stabilizes just above the stifle and the distal hand uses the calcaneus/tarsus as a hand-hold to perform the approximating oscillations.
- Stifle joint compressions in WB position: The distal hand stabilizes the paw on the ground with a hand-hold on the calcaneus/tarsus and the proximal hand performs the oscillations from just above the stifle toward the ground.

Frequency, Intensity, and Duration

Joint compression can be performed frequently throughout the day. Intensity is determined by the goal of the treatment. Duration is 1–2 minutes (Table 5.6).

Other Techniques

It is important to understand that there are numerous manual techniques used to address soft tissue and joint dysfunction. Each technique requires advanced study and training in order to be performed properly and safely. The following are a few such techniques.

Table 5.6 Depending on the goal of the treatment, the amount of force applied and the speed of the oscillations are varied.

Goal	Force of compression	Speed of oscillation
Pain	Light	Moderate to fast
Swelling	Moderate	Moderate
Proprioception	Light to moderate	Moderate to fast
Weight bearing	Moderate	Moderate

Joint Mobilization

As mentioned earlier, joint ROM consists of both osteokinematic and arthrokinematic motion. Joint mobilization is a technique used to treat arthrokinematic restrictions (Kaltenborn, 1989). Veterinary technicians and nurses are not trained in or approved to perform these techniques.

Arthrokinematic motion is passive, involuntary motion that occurs between two joint surfaces during osteokinematic motion. Arthrokinematic motion is assessed with a joint play assessment glide.

Arthrokinematic dysfunction is addressed with joint mobilization by gliding one joint surface on an opposing stabilized joint surface. Treatment parameters include: (i) type of mobilization (glide or traction), (ii) grade of mobilization (I, II, III, or IV), and (iii) direction of mobilization (cranial, caudal, dorsal, ventral, medial, lateral, etc.). The convex/concave rule is used to determine the proper direction of the treatment glide. This rule is based on the mechanics of a concave surface moving on a convex surface and vice versa.

Myofascial Release

Fascia is a three-dimensional web-like matrix of thin, tough connective tissue. It surrounds muscles, organs, bones, arteries, veins, and skin throughout the entire body (Salvo, 2012). The tissue is richly innervated with mechanoreceptors and these nerves can become sensitized and recruited as nociceptors (signaling pain).

Myofascial restrictions are caused by inflammatory responses, trauma, or surgical procedures. Fascia is a supporting structure that also aids in function. Myofascial release (MFR) is a soft tissue technique that specifically addresses fascial restrictions. Fascial restrictions are treated using either direct or indirect MFR techniques. The direct technique is a more aggressive technique. It involves placing all three planes of fascia into a stretched position ("restrictive barrier") until the sensation of a release is experienced in the therapist's hands. The goal is to create changes in the fascia by stretching the fascia and mobilizing adhesive tissues. The indirect

technique, on the other hand, is a gentler approach and more appropriate for the canine patient. In this case, all three planes of fascia are placed on slack (away from the barrier), allowing the fascia to "unwind."

Strain-Counterstrain

Based on the work of osteopathic physician Lawrence Jones, strain-counterstrain is a technique designed to reduce abnormal neuromuscular reflex within a tissue. His technique addresses areas of inappropriate proprioceptor activity that results in somatic dysfunction. On palpation, these areas are tense and tender and are referred to as "tender points." The technique is performed by passively placing the involved muscle or segment in a maximally shortened position. The position is held for 90 seconds. The goal of the technique is to reduce neuromuscular firing and to "reset" muscle tension. Expected results are a resolution of tension (muscle activity) and at least a two thirds reduction in pain on palpation within a treatment session. Strain-counterstrain is considered an indirect technique because its action is away from the restricted barrier (Basmajian and Nyberg, 1992).

Lymphatic Drainage

Lymphedema is a condition caused by damage to the lymph transport system (lymph nodes or lymph vessels). It is characterized by swelling of the limbs. Primary lymphedema is thought to be an inherited condition in humans and has been reported in a case study of two whippet siblings (Schuller *et al.*, 2011) and in draft horses (Affolter, 2013), whereas secondary lymphedema is usually the result of damage to the lymph system due to trauma, surgery, removal of nodes, or irradiation (Lee *et al.*, 2011).

The principles of manual lymphatic drainage (MLD) were considered as early as the 1800s by the founder of osteopathy, Andrew Still. His appreciation of the complexity of the lymphatic system stimulated a flux of further investigative work. MLD requires an extensive understanding of the anatomy and mechanics of the lymph system. The theoretical basis of MLD is that it: (i) stimulates

the lymphatic system via an increase in lymph circulation, (ii) expedites the removal of biochemical wastes from body tissues, (iii) enhances body fluid dynamics, thereby facilitating edema reduction, and (iv) decreases sympathetic nervous system responses while increasing parasympathetic nervous tone, yielding a non-stressed body-framework state (Vairo *et al.*, 2009). Treatment is often followed by the application of compression bandages (DeLisa *et al.*, 2005).

As a manual technique, MLD is a gentle technique in which rhythmic, repetitive, precise hand movements are used to affect the direction and speed of lymphatic flow (Willis, 2004). This technique is performed with light pressure (approximately 1–4 ounces (28–113 g)) in order to avoid collapse of the lymphatics. Treatment is initiated by “opening” the proximal node (that fluid is being drained toward) followed by more distal work, directing the fluid back toward the proximal node. This creates a path for the lymph as well as a suctioning effect to help draw the lymph toward the proximal node.

Manual Techniques as Home Exercise Program

One of the most important elements of a successful rehabilitation program is the home exercise program (HEP). To maintain and/or progress gains made in the clinic, the owner must be compliant with a home program. Thorough owner education is the key to compliance. See more about the home exercise plan in Chapter 6.

Owner education should include:

- an explanation of the injury or condition in layman’s terms,

- an explanation of the purpose of each technique/exercise as it relates to resolution of the condition and improved function for the patient, and
- an explanation of how to identify a red flag or a negative response to the treatment.

Once the owner understands the relevance of the HEP, instruction in performing the program can take place. There are three components to instructing an owner in a HEP.

- *Demonstration:* A visual and verbal demonstration should be performed. This includes a verbal explanation of handling skills, hand placements, and specific nuances.
- *Practice:* The owner should practice all techniques/exercises while in the clinic until performed safely and effectively.
- *Take-home instruction:* Owners should be provided with written instructions and pictures of each exercise or technique, including frequency, intensity, and duration of each element. Additionally, video recordings done by the owner on phone or video camera can be extremely helpful.

Conclusion

The goal of veterinary physical rehabilitation is to restore maximal function. In this chapter, we discussed one element of a rehabilitation program—manual techniques (AARV, 2016). These techniques promote tissue health and mobility, laying the groundwork for normal function. With the appropriate use of manual techniques in combination with an individually designed therapeutic exercise prescription, effectiveness and efficiency in the rehabilitation process can be achieved.

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6

Home Exercises

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CHAPTER MENU

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Introduction

Client involvement in a rehabilitation program may be minimal, or the therapy may be performed entirely by the client in the home environment. To design the ideal program, both the client and the patient must first be assessed. The client must be committed to the plan, and have the mental, emotional, and physical capability to work with the patient. The rehabilitation team must work with the client to create a program that is effective and fits with the client's capabilities and resources. The therapist must ensure that the client can correctly perform the therapies, and the patient should be assessed regularly until the therapeutic goals are reached.

Assessing the Client

Level of Commitment

Several factors can affect a client's level of commitment to a rehabilitation program. It is important to understand some of that factors that may impede the client from participating in the therapy. Once the rehabilitation team is aware of potential factors that may reduce client compliance, they may be able to adjust the program to improve the client's ability to commit.

Probably the most important factor for client compliance is whether they value rehabilitation as a treatment. The client's previous personal experience with exercise or with physical therapy can have a positive

or negative effect on their attitude toward rehabilitation for the patient. The client's cultural background may also affect their beliefs in the value of therapy. A basic understanding of the patient's health condition and how rehabilitation can help is also important. Client education by the rehabilitation team can make a significant difference in the client's understanding of the value of rehabilitation, which may help motivate them to follow a program.

Financial considerations can also affect client compliance because clients must be willing and able to pay for a rehabilitation program. Sometimes financial considerations will affect the type of program the patient is entered. If clients have minimal financial constraints, they may be able to take advantage of extensive in-house therapy. For many patients and clients, it is appropriate to consider some in-house therapy along with home exercises. However, if funds are limited, the client may request a program that is strictly carried out at home with minimal intervention on the part of the rehabilitation team. A frank discussion of the cost of rehabilitation enables the rehabilitation team to develop a program that is affordable for the client to carry out. In some cases, the less expensive option may be less ideal for the patient. However, this may be preferable to the client electing to pursue no therapy because it is unaffordable.

Time constraints may also affect the degree to which the client adheres to a program. If the rehabilitation program recommended is too time intensive, the client may become overwhelmed and entirely abandon it in frustration. It is important to have a non-judgmental discussion to establish how much time the client can realistically commit to the program each day. Once this has been established, a program can be designed to maximize the available time.

The client's understanding of the value of rehabilitation, along with any financial or time constraints should all be discussed prior to creating the rehabilitation program. Any factors that may be barriers to commitment

must be considered when designing the program, because the "ideal" program will not be effective if the client is unable to remain committed to it.

Physical Abilities

Rehabilitation can be very physically and mentally demanding for the client or the animal handler. The client's general physical abilities should be assessed during the initial discussion about the rehabilitation program. If the client volunteers information about personal medical conditions or physical limitations, discuss how the program may be designed to adapt to these conditions. If the client does not volunteer any information, the therapist must decide whether it is appropriate to initiate a conversation pertaining to the client's physical capabilities.

The therapist should carefully observe the client during any training sessions where the client is demonstrating proficiency in handling the patient. The client may be required to get down on the ground with the patient for some exercises and may need to transition from a squatting to a standing position frequently. In some cases, the client may need to lift and support the patient. In addition, some exercises may require a certain level of agility or speed on the part of the client, if the patient is to perform correctly. Exercises that are complex or require precise timing may be too difficult for some clients to understand and perform well. If the therapist suspects that the client may have physical or mental difficulty with certain exercises, they may need to be modified or dropped from the home program.

There are situations where multiple handlers are required to have the patient perform the treatments correctly. This most commonly occurs in large dogs that are non-ambulatory or have severe physical impairments. In these situations, the client must be able to identify other individuals in the home environment who will be able to participate in the patient's care.

It is very important for the client to be physically fit for them to perform rehabilitation on their pet. Many home exercise programs include a lot of walking or stair climbing. If clients have been sedentary or have any potential risk associated with an increase in activity, it should be recommended that they consult their physician before beginning the rehabilitation program for their pet. The goal is to improve the patient's physical ability, not decrease the client's! However, sometimes the patient's need for exercise is helpful motivation for the clients to improve their own physical fitness.

Animal Handling Skills

The animal handling skills of the client can affect the design of the rehabilitation home program. Knowledge of basic obedience commands is nice but not necessary. However, the client must be able to control or redirect the patient's behavior. If the client is unable to control the patient's movements they may be performed ineffectively, or even worse, could result in further injury to the patient.

As clients understand more about animal training, they are better able to motivate or alter patient behaviors. Although this is not essential for passive types of therapy, it becomes more important for active exercises, particularly if the movements become more complex. For example, controlled leash walking can be performed with minimal handling skills (although this can vary with the patient's personality and level of training). However, having a patient perform sit-to-stand exercises requires a higher level of control that can be made easier if the client can shape the patient's behavior.

The client should also have a basic understanding of animal behavior. Some clients may need guidance to understand body language. This is particularly important if the patient may be inclined to bite or scratch, so the client can avoid becoming injured during a therapy session. The client should also be instructed to interpret signs that the patient

may be painful. This is helpful because additional therapies may be indicated if the patient is painful. In addition, clients are fearful of creating pain and become reluctant to perform the therapy. If they can more accurately assess the patient's pain, they are more likely to perform appropriate therapy.

Clients may benefit from some instruction to improve their animal handling skills. It is important not only to explain handling skills to the client but also to show them physically, and then have them demonstrate proficiency. In this way, the therapist can identify potential problems in the animal handling techniques.

Client–Patient Bond

The relationship between the client and the patient plays a big role in successful rehabilitation. The client's emotional bond often provides significant motivation for the client to persist in helping with the recovery process. In addition, the shared bond can often enable the client in motivating the patient to perform desired activities. Clients may understand what toys or treats best motivate their pets, as well as which activities they particularly enjoy. A more intimate knowledge of the patient's personality may give clients an advantage, because they may be able to “think outside the box” to motivate the patient to perform an exercise or movement.

In some cases, the client relationship may hinder the rehabilitation program. Some clients may be reluctant to push the patient to do any activity when they encounter resistance. In addition, some clients may be concerned that they are going to create pain or further injury. These clients must be educated so they understand what needs to be done, and gain confidence that their efforts are necessary and beneficial.

Clients need to be able to notice any subtle changes in the patient. They need to be instructed to take note of any physical changes that are either seen or felt by physical palpation and take note of any behavior change that arises during their in-home treatments. And as previously mentioned,

the clients must also be able to watch for signs of pain, and distinguish it from fatigue or stress. Some clients are very skilled at understanding their pet's personality and behaviors, while others may need some guidance and education related to typical animal behaviors.

Rehabilitation Goals

Rehabilitation goals should be discussed early in the assessment process. The rehabilitation team can help establish realistic end goals, based on their knowledge of the condition and prognosis. The team can also suggest whether the goals are likely to be achievable with the patient receiving home care only. In many cases, a combination of in-house therapy and home care, or extensive in-house therapy may be more appropriate.

Client expectations should also be explored during this discussion. It is important for the therapy team to know what the client anticipates and desires as the outcome. Do they want their pet to return to full working duty or do they just want it to be able to climb up on the couch again? If there is a disparity between what the client expects and the probable outcome, this should be recognized before beginning the therapy program. Although it is beneficial to be optimistic, a realistic outcome needs to be discussed.

Assessing the Patient

Motivation

Motivating patients to work can be one of the most difficult aspects of rehabilitation. There are several factors that can affect individual motivation. Patients that are painful, fatigued, or stressed may be distracted or reluctant to be active. Very young patients may have a brief attention span, while geriatric patients may have cognitive impairments that reduce their ability to follow directions (Salvin *et al.*, 2010). Patients that have already undergone some type of training may be better able to

understand what the handler or client is trying to get them to do. In addition, some patients are more adept at learning what behavior is expected of them. Although popular opinion may be that certain breeds are easier to work with than others, every individual is different.

Using food or treats as motivation is one of the easier and more popular ways of getting the desired behavior. However, not every patient is food-motivated. When treats are not effective, various toys may be tried. In some cases, many different types of toys must be tried until one is identified that is particularly motivating to the individual. Some patients respond best to positive verbal feedback from their owners. Teaching an owner the appropriate time and type of praise during exercise is important to ensure the patient understands the desired behavior. Often it is necessary to be innovative and to try different techniques before discovering what works best for the patient. Regardless of what typically works best for the therapist, always remember that it will not work for every patient.

Training

It is important to consider the patient's level of training. If patients do not already know basic obedience commands such as "sit," then it may be more difficult to have them perform sit-to-stands. The therapist may be able to train basic skills as the therapy progresses. However, if the client is performing home therapy, they may be unwilling or unable to train the patient. The complexity of the exercises must be chosen with the patient's level of training in mind. If the patient is not trained to do a certain command, it is still possible to lead them through the movement or shape the behavior, with the right type of motivation. Bear in mind that the client must possess or learn training skills if the therapy program is being done at home.

Some patients already have a high level of training. This can make it easier to design a varied therapy program. However, this type

of patient may lose focus if the therapeutic activities are very simplistic and repetitive. For this type of patient, it may be necessary to challenge them not only physically but mentally as well.

Goal of Exercise Program

The goal is to return the patient to the highest level of function as is possible. To do this, the therapist must understand which exercises are beneficial and which are contraindicated. To meet the goal it is sometimes best to focus on accomplishing short-term goals or individual exercises. It is important to remember that the patient must be evaluated at regular intervals throughout the rehabilitation program. It may be necessary to adjust goals or the expected time the goals are reached as the patient progresses through the program.

Assessing the Resources

Environment

Factors in the home environment will influence the development of the therapeutic exercise program. As the activity level is increased, walking up stairs or hills may be recommended, if available. A pond, lake, hot tub, or swimming pool may be recommended for water therapies such as walking in water or swimming. However, these may not be appropriate to include in the therapeutic plan if the client has limited access to these resources or is unable to use them safely.

The patient's environment may also need to be controlled to prevent further injury. Most patients need to be housed indoors, and often need to be confined to a relatively small area when they are unsupervised. This is typically best accomplished by use of a kennel or a small room. Early in the rehabilitation program it may be undesirable for patients to be traversing flights of stairs, and this can present a challenge if they are in the habit of sleeping in an upstairs bedroom. In

addition, patients who are recumbent need a padded area to rest. If the home has hardwood or tile floors, weak patients may need non-skid boots or assistance to prevent slipping. The dog door may also need to be temporarily inactivated to limit unsupervised access to the yard. Appropriate activity levels should be discussed to ensure the patient is prohibited from doing too much too soon.

Equipment

Knowledge of the client's available equipment is just as important as understanding the home environment. The equipment that the client has or that they can make will heavily impact the structure of the home exercise program. Most therapeutic exercise equipment can be purchased inexpensively at a local department store, but clients who are crafty and willing can make some of the equipment. Another option may be for the rehabilitation business to loan, rent, or sell equipment to clients.

A basic rehabilitation home program can often be created with minimal to no equipment. If the patient needs to reach a high level of function, particularly if they need to return to sports-specific or work-specific activity, some specific exercise equipment may be needed. However, for this type of patient, clients usually have access to necessary training equipment.

Training the Client

Initial Instruction of Exercises

The initial instruction with the client must be very detailed. This begins with a handout that explains the home program. A home exercise program should not just list the exercises but also give detailed descriptions and photos of the exercises and explain how to perform them. The client is given so much information during the discharge that it is easy for them to forget what was said or lose a piece of paper. Providing a video of the

exercises is also very helpful. The correct form can more easily be conveyed by video as compared to written instructions. Clients may be overwhelmed by all the instructions, so it is helpful for them to have information that they can refer to once they get home.

Providing verbal and written instructions is essential, but these alone are not sufficient. The therapist should demonstrate the correct techniques with the patient. The client should then practice the techniques with the therapist's guidance. This allows the therapist to ensure that the client understands the exercises and can perform them correctly with the patient. This session also gives the client opportunity to ask questions, and the practice activities improve their self-confidence. A client who has demonstrated appropriate technique, and who is confident that they can perform the therapy, is more likely to adhere to the program.

Finally, if any assistive devices are sent home, the therapist should ensure that the client is completely comfortable with how they work and how to apply them. Slings, carts, and braces should be evaluated by the therapist or veterinary team to ensure they fit the patient properly. It is best to have the client practice applying and removing the device several times, until the client and therapist are confident that the client can use it correctly. The client must also be advised to watch for complications. Specific regions of the body where an apparatus may pinch or rub should be pointed out, and specifics of when to act should be given when "any issue more than hair loss needs to be addressed." In general, the client should be instructed to call immediately if they notice any potential problems and to avoid the use of the device if possible until examination can be performed.

Education Regarding the Patient's Condition

It is very important for the client to be aware of the patient's condition. Client education on pain assessment is key in ensuring that

the patient's comfort level is adequately addressed. In educating clients about pain, they must be told the signs to watch for during exercises that may indicate a problem. Signs of pain can include vocalizing, excessive panting, dilated pupils, limping, restlessness, and change in behavior or personality (Gaynor and Muir, 2015).

Clients also need to know what complications may occur from any of the recommended treatments, physical or pharmaceutical. The potential complications will vary depending on the initial problem and the patient's condition, and must be identified by the rehabilitation team. The potential complications should be explained to the clients, and they should also be given a written description of potential signs. Clients should also be given guidelines that outline when an exercise should be stopped and when to seek additional help. In addition, the clients should be given an idea of the rate of progress to expect as they progress through the program.

Follow-Up

The follow-up evaluations are just as important as the initial exam. These evaluations are used to determine whether the patient is improving as expected or new problems are noted. If the patient is not improving as expected, the reason for this must be evaluated. It is possible that the initial goals were overly optimistic. It is also possible that the client is having difficulty following the rehabilitation program, or the patient may be experiencing complications. Regardless, the rehabilitation program may need to be modified to address any problems.

A phone call is one way to see how the client is coping with the exercises at home. Weekly phone calls allow the therapist to assess whether the client is comfortable with the prescribed exercises and to obtain an update on the patient's progress. Bear in mind that the client's self-report of patient progress may be inaccurate. In addition, a client may be unaware that they are performing

therapy incorrectly. Therefore, it is important to have the client repeat the exercises after they are shown how to perform them during the initial assessment. The therapist may also help motivate the client by phone if they are having difficulty performing the exercises or becoming frustrated with lack of progress.

Ideally, the patient should be re-evaluated roughly every 4 weeks depending on the problem. Some will need to be seen sooner and some later. If the patient is also receiving in-house therapy, a re-evaluation can be scheduled during one of these visits. The patient should not be released to full activity until a final evaluation is performed.

Home Exercises

Thermotherapy

Application of cold packs is beneficial after any surgery because it helps reduce inflammation and aids in pain control (Cameron, 2003). Cold packs may also be applied after exercise to reduce inflammation. This may be especially beneficial for patients with arthritis, because the cold can reduce the

pain and inflammation of arthritic joints. A cold pack can also be applied to an area that has been traumatized, to reduce hemorrhage and inflammation (Cameron, 2003).

Several different forms of cold packs are available for use. Commercially available gel packs are nice because they remain pliable even when frozen, can be easily disinfected, and usually can double as a hot pack (Figure 6.1). Crushed ice can be placed in a sealable plastic bag to create an ice pack. A small amount of water may be added to help eliminate the dead space. Ice can be placed in a pillowcase or moist towel, although this can be messy. Using a bag of frozen vegetables has been recommended for emergency situations. This is a less effective method because of the dead space between the vegetables, so is not recommended for use during therapy sessions. An ice slush can be made by mixing four parts water to one part rubbing alcohol in a sealed bag and placing in a freezer. Note that this alcohol–water mixture performs at a lower temperature than crushed ice. Therefore, this type of cold pack should never be applied directly to the skin because it carries an increased potential for causing cold-induced injuries (Cameron, 2003).



Figure 6.1 Commercial gel packs are relatively inexpensive. The rehabilitation facility may sell them to the client, or they can be purchased at a local drug store.

Before applying a cold pack it can be wrapped in a cloth or thin towel to protect the skin and keep the cold pack clean. For maximal cooling effect, the towel can be moistened. However, when using a true ice pack or alcohol slush pack, the cooling is more intense, so the towel should be dry. The cold pack should be held firmly in place or secured with an elastic bandage to ensure good skin contact. The application time should be at least 10 minutes to effectively reduce pain and swelling, and may be repeated every 1–2 hours (Cameron, 2003). It may not be necessary to treat that often, and may be impractical for the clients. A typical recommendation is to apply the cold pack for 10–20 minutes (Dragone *et al.*, 2014), 2–4 times daily, or after each therapy session.

Heat therapy can also be used in a home program. Some of the effects of heat are peripheral vasodilation, increase in metabolic rate, relaxation of muscle spasms, improvement of soft tissue elasticity, and reduction in pain. Indications to use heat would be to help reduce pain and muscle spasms, to enhance stretching, or to warm the tissues before activity. Use of local heat therapy is contraindicated in the presence of active bleeding, acute inflammation or infection, swelling or edema, neoplasia, or impaired sensation to the area (Cameron, 2003). This means that heat should not be applied near a surgical incision during the first few postoperative days when there is inflammation associated with the surgical procedure! A hot pack encourages dilation of blood vessels, resulting in increased blood flow and metabolic activity to the area (Cameron, 2003); this would be undesirable if there is postoperative inflammation.

The most commonly used hot packs are sacks or bags filled with beans, rice, cracked corn, bentonite (a hydrophilic silicate gel), or other inert materials. As mentioned above, the same gel packs can be used as either cold packs or hot packs. A hot pack can also be created by heating a damp towel or bag of water in the microwave; however, these

methods may result in less effective heating. Although hot packs can be beneficial, bear in mind that heat from a hot pack only penetrates a few centimeters beneath the tissue surface.

The hot pack should be checked against a person's skin to ensure that it does not feel too warm for the patient. A towel or cloth may be placed between the hot pack and skin if there are any concerns about overheating the skin. In addition, the skin should be observed intermittently for signs of excessive redness. In addition, animals with normal sensation should be monitored for evidence of discomfort. Hot packs are generally applied to an area for 15–30 minutes. This may be done 2 or 3 times daily, and may be done as a warm-up prior to exercise or stretching.

Passive Range of Motion

Passive range of motion (PROM) is performed by moving a joint within its available range of motion, using an external force. The benefits of PROM are that it maintains range of motion, helps prevent joint contracture, reduces pain, and improves synovial fluid production and diffusion (Millis and Levine, 2014). It is important to remember that PROM does not prevent muscle atrophy, nor does it improve strength or endurance. PROM can enhance blood and lymphatic flow, but not as effectively as active range of motion does (Kisner and Colby, 2007). Range of motion exercises are contraindicated if the movement could cause further injury or instability. For example, a tenuous fracture repair could be compromised by excessive movement, so it is very important to discuss the therapeutic plan with the surgeon before beginning therapy. In most cases, PROM is beneficial if performed at a slow, controlled speed and within a comfortable range of motion for the patient.

It is best to perform PROM in a quiet room with as few distractions as possible so the patient can relax. The patient should be in lateral recumbency with the affected leg up.

If the patient is aggressive, it may need to be muzzled. If the patient seems to be painful, the pain management protocol may need to be modified prior to performing PROM. In some cases, two people are needed to perform PROM—one to restrain or calm the patient and one to do the exercises.

When performing PROM on a postoperative orthopedic patient, it may be most efficient to focus on the affected joint; however, it is important to maintain normal range of motion in all joints. When performing PROM on a neurologic patient with a spinal cord injury, PROM should be performed on all major joints. For example, if the patient is paraplegic, PROM should be performed on the joints of both hind limbs, and if the patient is tetraparetic PROM should be performed on the joints of all four limbs.

It may be necessary to spend a few minutes gently massaging the affected limb to promote further relaxation, and gradually move toward the affected joint. Some patients do not like their feet being touched so it may be better to start proximally and move slowly, rather than grabbing hold of the patient's foot. Motions should be slow and controlled. This is not a race! One hand should be positioned to support the limb proximal to the joint, while the other hand supports the limb distal to the joint. The entire limb should be supported with the joints in neutral positions to prevent any excessive joint stresses. The two hands are slowly moved to gently flex the affected joint, while the other joints are allowed to maintain neutral positions. The joint is flexed as completely as possible until some resistance is met or until the patient displays signs of discomfort. Discomfort may be displayed by muscle tension, pulling the limb away or turning the head toward the affected area. With hands in the same supportive positions, the joint is extended fully, stopping before the patient displays signs of discomfort. An alternative method is to support the entire limb and flex and extend all the joints by moving the limb in a way that mimics an exaggerated walking step, or "bicycling," movement. This method of

PROM is more appropriate when the patient is nearly using the leg actively. Remember that the stifle and hock positions are related, so maximum flexion of one of these joints requires simultaneous flexion of the other. When done correctly, PROM should not be a painful experience nor should it cause any signs to worsen.

The most appropriate treatment prescription will vary with the condition. However, for most postoperative conditions, 15–20 repetitions (for each joint), 2–4 times daily is adequate (Millis and Levine, 2014). PROM is typically discontinued when the patient can use the leg, and flex and extend the affected joint voluntarily. Although the patient may not be completely normal at this point, the focus of therapy changes to promote active range of motion exercises.

It is important to show the clients how to correctly perform PROM so that it can be done correctly and safely at home (Figures 6.2 and 6.3). If the client struggles to master the technique or the patient is uncooperative, the therapy team must decide whether the client should persist in attempting to do the PROM. It is possible that the potential risk of injury (to the patient or the client) outweighs the anticipated benefit of PROM. However, most clients can learn how to perform effective PROM. In these cases, it is still recommended that they demonstrate their technique during patient re-evaluations because they may not realize when their technique is suboptimal. Most clients enjoy learning this skill and it can be empowering for them to feel they are contributing to the successful recovery of their pet.

Stretching

Various stretching techniques may be used to improve joint range of motion and extensibility of periarticular tissues, muscles and tendons. Passive stretching is not the same as PROM. PROM activities occur within the unrestricted range of motion of the joint, while stretching moves a restricted joint beyond the available range of motion to



Figure 6.2 Every home therapy should be demonstrated to the client with their pet.



Figure 6.3 After receiving instructions, the client practices the home exercise with feedback from the rehabilitation staff.

elongate the soft tissues (Kisner and Colby, 2007). Static stretching is similar to PROM from the standpoint that the patient is not actively involved in the movement.

A static stretch is performed using a similar technique as for PROM. The patient should be

as relaxed as possible, and the limb is held and supported on either side of the affected joint. The joint is then moved to the end of its available range. This may be flexion or extension, depending on where the joint restriction is located. There is limited evidence from which

to determine the most effective stretching protocol. In healthy adults, stretching for 30 or 60 seconds was more effective than 15 seconds or less (Bandy *et al.*, 1997). There is reportedly no increase in muscle elongation beyond 2–4 repetitions of static stretching (Page, 2012). The optimal frequency and duration for static stretches is still being researched. Static stretching is typically a little uncomfortable for the patient, and can be more challenging to

perform than PROM. In addition, there is a higher risk of causing injury to the patient. For these reasons, static stretching may not be the best choice for a home program.

Active stretches can be performed by coaxing or luring the patient to move into positions that accentuate joint flexion or extension, which stretches the associated soft tissues (Figures 6.4). Since the patient is controlling the motion, the risk of overstretching is

(a)



(b)



Figure 6.4 The patient can be encouraged to stand in a position that results in active stretching of specific joints. (a) Active stretch on a disc – little dog. (b) Active stretch on a bosu – bigger dog.

(a)



(b)



Figure 6.5 Exercises can be prescribed based on what equipment the client has available. Clients can be instructed how to perform active stretching with no equipment. However, it is important to avoid training the pet to perform any behaviors that may be undesirable. (a) Active stretch on a person – little dog. (b) Active stretch on a person – bigger dog.

relatively low. These stretches are often performed by use of a toy or treat to lure the patient into various positions, hence they may be termed “cookie stretches.” One example of an active stretch is to coax a dog to stand with its hind legs on the floor and its front legs on a stair step (which step depends on the size of the patient and degree of stretch desired). This could be used to encourage hind limb extension, promoting stretch of the hip and stifle flexors (Figure 6.5).

The classic “cookie stretch” can be used to promote stretching of the cervical and thoracic spine. It is also a type of weight-shifting exercise that can simultaneously help improve balance, and limb and back strength. This is performed by having the patient stand on a surface with good traction. The cookie is

used to lure the patient’s head to one side, and then the other. Luring the patient’s nose to the shoulder will focus on stretching of the neck, while luring to the hip or hock will focus stretching of the trunk. The nose may also be lured up and down to stretch the ventral and dorsal aspects of the neck. The patient can be enticed to hold the stretch by delaying delivery of the treat while the patient is maintaining the end range position. Although this is a straightforward exercise, it can be challenging for some clients to master the luring techniques so the patient remains in a standing position without moving the feet. The therapist must decide on an individual basis whether this type of stretch is indicated, and if so, whether the client will be able to perform it correctly.

Massage

Massage is the manipulation of soft tissues, and several different styles have been developed. The basic concept is to apply pressure and friction to the patient's skin. Massage can improve lymphatic flow, and one of the indications is to reduce limb edema. Massage helps reduce anxiety and pain, but there is limited scientific evidence to support other potential benefits (Sutton and Whitlock, 2014).

Clients often enjoy giving massages to their pet. The pet may become more relaxed as a direct result of the massage, and it often appears to improve the bond between the client and patient. Massage may be performed prior to a therapy session to calm the patient. Massage may also be performed following exercise, particularly to reduce muscle spasms and pain.

The type and extent of massage may depend partly on the client's level of comfort and skill. The first step in performing a massage is to have the patient lie in lateral recumbency in a quiet room. The client may be instructed in basic stroking, effleurage, and petrissage techniques (see Chapter 5). The massage may be general (for general relaxation and pain reduction), or focused on tissues of the affected area. The initial massage may be supervised by the therapist to ensure that the client is using appropriate techniques. The massage should not cause pain or discomfort.

Weight Shifting

Weight shifting exercises are relatively simple activities that most clients can easily learn to perform. This type of activity can be done for patients who can stand, but who need to work on improving balance and fine motor control. Shifting of the body increases mechanoreceptor firing in the joints and soft tissues, which increases feedback to the central nervous system regarding body position. This feedback results in increased muscle recruitment to stabilize the body (Deliagina

et al., 2014). Weight shifting may also be performed to encourage weight bearing on a limb. The simplest type of weight shifting is done with the patient standing (with assistance if needed), while someone gently pushes the patient from side to side or front to back in a rhythmic fashion.

For example, the client may place one hand on each shoulder or hip and gently sway the patient from side to side. Making the movements larger or more rapid will increase the difficulty of the exercise. Another method of weight shifting is to use the "cookie stretches" as previously described. Weight shifting activities can be made even more challenging by having the patient perform them while standing on an air mattress or couch cushion. There are many other types of weight shifting exercises, but they require equipment that most clients do not have at home. In addition, these are more complex and require more skill to do correctly, so may not be appropriate for many clients.

For more details about weight shifting, see Chapter 19.

Assisted Walking

Patients that are weak or non-ambulatory may need assistance to stand and to walk. The patient should be fitted with the most appropriate sling to be used by the client at home. It is important to instruct the clients on proper techniques for lifting and assisting patients, to preserve client health. In some cases, the patient may be fitted for a cart, particularly if the dysfunction is anticipated to be prolonged or permanent.

Clients with patients that spend most their time recumbent must be instructed on basic nursing care to prevent urine scald and decubital ulcers (see Chapter 11). The clients must also be alerted to watch for foot abrasions in patients that have neurologic deficits or abnormal postures. They should seek veterinary care as soon as they notice any reddened skin beginning to develop. Foot abrasions can often be prevented by boots.

Controlled Leash Walking

Slow leash walks are typically a key component to most rehabilitation programs, particularly early in the rehabilitation process. Slow leash walking is indicated for patients who have lameness, weakness, or proprioceptive deficits. The authors have observed that the patient is more likely to use each leg in the proper gait sequence and more likely to bear weight on a lame leg when encouraged to walk slowly. It may help to praise lame patients whenever they touch the affected foot to the ground.

Instructions for the clients must be very clear and detailed. For maximum control, the leash must be a short lead (1–2 meters long), and not a retractable leash. Ideally, the patient should be walked in a normal “heel” position. If the client–patient team lacks obedience training, the patient may walk a few steps ahead or to the side of the client, but it must be as controlled as possible. Retractable leashes are strongly discouraged because the tendency is for the client to relinquish control of the patient, allowing activity that is inappropriate or too vigorous.

The appropriate duration and frequency of the walks varies with the patient’s condition. A typical postoperative program might start with 5-minute walks, 2–4 times daily. Over time the duration of the walks may be gradually increased. In general, the duration or frequency of the walking sessions should only be increased after the client instructed to do so by the rehabilitation team. A guideline that can be used to increase cardiorespiratory conditioning is to increase the duration of the walks by 10–15% each week (Millis *et al.*, 2014a). However, this guideline may not be appropriate for patients recovering from orthopedic or neurologic injuries or surgeries.

Balance, strength, and active range of motion exercises may be incorporated into the patient’s walking program. During each walking session, the patient may spend a portion of the time walking on uneven or

unstable surfaces, such as grass, sand, or snow, if available. Walking over different terrain may challenge the patient’s balance, and it can also encourage flexion of the limb joints. Joint flexion may also be enhanced by exercises in which the patient steps over objects, such as a garden hose, broomstick, PVC pipe, or rungs of a ladder placed on the ground (Figure 6.6). Clients who have working or sporting dogs may have agility equipment, such as cavaletti rails that they can use. Inclines, declines, or stairs may also be integrated into the walking program, depending on what is accessible to the client. Walking down stairs is particularly beneficial for increasing range of motion of the stifles and hocks (Millard *et al.*, 2010), while walking up ramps promotes range of motion of the forelimb joints (Carr *et al.*, 2013).



Figure 6.6 Step-over exercises can be prescribed using materials that can be obtained cheaply and easily by the client.

Strength Exercises

A walking program will improve strength of the limbs. Walking up hills or stairs may be used to focus on strengthening of the hind limbs. The client must understand the importance of moving slowly to discourage the patient from hopping or skipping up the steps. The client must also be able to recognize signs of fatigue, and avoid pushing the patient too quickly. Hills or stairs can be challenging for the cardiorespiratory endurance and muscular strength, so these exercises should be gradually introduced into the program, particularly if the patient is deconditioned.

Sit-to-stand exercises can also be incorporated into a home program to improve strength and range of motion of hind limbs (Millis *et al.*, 2014a). It is best to have the client include some sit-to-stand exercises after 5 minutes of walking to warm up the muscles. Although sit-to-stand exercises look simple, they can be difficult to perform correctly. The patient must sit and stand symmetrically, without leaning to one side or favoring one leg. In the authors' experience, patients often tend to abduct the affected limb to avoid fully flexing it during the sit or fully bearing weight during the transitions. The client must be instructed to watch for this method of "cheating." It may be prevented by positioning the patient with the affected leg next to a wall or other object that prevents lateral movement of the leg during the exercise. It may be most efficient to start by including 5–10 sit-to-stand exercises once or twice daily, during the walking sessions (Millis *et al.*, 2014b). The number of daily sessions and repetitions can be increased over time as the patient's strength improves.

Crouched walking is another relatively simple exercise to improve limb strength and range of motion. One technique is to have the patient walk through a child's play tunnel or dog agility tunnel. However, the tunnel must be small enough that the patient cannot stand up inside it. This type of equipment may not be available to the client; however,

other objects may have the same effect. For example, the client may have a chair or table that the patient could walk under. Another convenient and inexpensive option is a telescoping shower rod. It can be placed in a doorway or hallway at an appropriate height to force the patient to flex the limbs to walk under it.

Sport-Specific Exercises

The end goal of the rehabilitation program is that the patient returns to full function. In many cases, this does not involve a high degree of athleticism. However, the goal for some patients is to return to a competitive sporting activity or to return to strenuous work. For these patients, the clients typically have a good understanding of the type of strength, endurance, and agility that is required. These clients also tend to have more experience with animal training and access to specific training equipment.

The final portion of the rehabilitation program for these patients should focus on building the specific skills and endurance needed for that individual. The rehabilitation team can work with the client to design a program that is gradual, yet progressively improves patient function. Although this type of client tends to be very observant, it is still important to reassess the patient at regular intervals to ensure they are meeting retraining goals and not sustaining any injuries or setbacks.

Conclusion

The therapist must assess both the client and the patient carefully before creating a home exercise program. The therapist and the client must establish realistic outcome goals. Clients must be committed to the therapeutic plan. The therapist must determine which therapies are realistic for the clients to perform at home, considering their available resources and their animal handling skills.